

Unit - 2: Damped and Undamped Vibrations

Pg-1

Topics : Derivation for spring mass system ; Methods of Analysis ; Natural frequencies of simple systems, Springs in series and parallel ; Torsional and Transverse vibrations - Problems. Derivation - Over, Critical and under-damped system ; Logarithmic decrement and problems.

Free vibration : When no-external force is applied to the system, but the elastic system vibrates because of inherent forces, it is called as free vibration.

Undamped Vibration : If there is no dissipation of energy due to friction or resistance in a vibrating body, it is known as undamped vibration.

A simple case to analyse the motion of a vibrating system is to consider free vibration with single degree of freedom spring-mass-undamped structure.

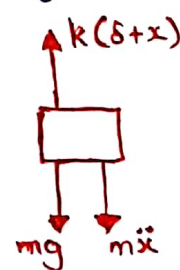
Derivations for spring Mass systems

Theoretically there are many methods available to obtain equations of motion for a single degree of freedom system, but in this unit, let us discuss three important methods used.

[1] Newton's method

Case 1: Spring Mass system in vertical position

Consider a vibrating system and having one coordinate to represent the motion of the system. For simplicity let us assume the vibration to have rectilinear motion along the axis of the spring as shown in figure



Free body diagram

where

$k \rightarrow$ Spring of constant stiffness

$m \rightarrow$ mass of the system

Because of the mass acting vertically downward, the body is displaced vertically downward.

The equilibrium position is called static equilibrium.

From the figure (2), it is clear that the gravitational pull (W) is balanced by a force of spring, therefore.

$$mg = W = k\delta$$

[$\because$ Stiffness (k) = force/displacement]

where $\delta \rightarrow$ Static deflection of the spring

Because of the self weight, the mass is displaced from its equilibrium position by a distance ' x ' (free vibration) and then released. So after time ' t ', we have.

$$\text{Restoring force} = W - k(\delta + x)$$

$$m\ddot{x} = W - k\delta + kx$$

$$m\ddot{x} = -kx$$

$$[\because W = k\delta]$$

$$\therefore \boxed{m\ddot{x} + kx = 0}$$

$\rightarrow [1]$

where $\ddot{x} \rightarrow$ Acceleration of mass ' m ' $= \frac{d^2x}{dt^2}$

Let us assume the vibrating motion to be Simple Harmonic, therefore the solution is $x = A \cos \omega_n t + B \sin \omega_n t \rightarrow [2]$

A and $B \rightarrow$ Constants which can be found from initial conditions
 $\omega_n \rightarrow$ Circular frequency of the motion.

Substituting ~~for~~ ' x ' in equation (1), we have

$$- \omega_n^2 (A \cos \omega_n t + B \sin \omega_n t) + (k/m)(A \cos \omega_n t + B \sin \omega_n t) = 0$$

$$\text{Since } (A \cos \omega_n t + B \sin \omega_n t) \neq 0$$

$$\therefore - \omega_n^2 + k/m = 0$$

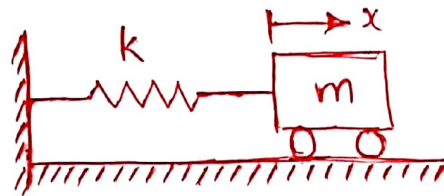
$$\therefore \omega_n = \sqrt{k/m} \text{ rad/s}$$

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The frequency of vibration is

$$f_n = \frac{\omega_n}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

Case 2: Spring-mass system in Horizontal Position.



Consider a spring mass - system having spring stiffness denoted by 'k' and having mass mounted on a frictionless roller.

According to Newton's second law

mass $\times$ acceleration = resultant force on the mass



Free body diagram

$$\therefore \boxed{m\ddot{x} + Kx = 0}$$

Note: Frequency of a given system is same whether it vibrates in a horizontal (or) vertical position.

[2] Energy Method

According to this method, in a conservative system, the total sum of the energy is constant.

- In a vibrating system, the energy is partly potential and partly kinetic
- Because of the velocity of mass, the system has Kinetic energy (K)
- Since there is elastic deformation in spring, potential energy (P) is stored in the spring.

According to law of Conservation of energy

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$$P + V = \text{Constant}$$

Differentiating the above equation w.r.t time, it is zero.

$$\frac{d}{dt}(P + V) = 0$$

$$\text{Since } P = \frac{1}{2} m \dot{x}^2 \text{ and } V = \frac{1}{2} K x^2$$

$$\therefore \frac{d}{dt} \left(\frac{1}{2} m \dot{x}^2 + \frac{1}{2} K x^2 \right) = 0$$

$$m \ddot{x} \dot{x} + K x \dot{x} = 0$$

$$\boxed{m \ddot{x} + K x = 0}$$

[3] Rayleigh's method

According to this method, it is assumed that the maximum Kinetic energy at the mean position is equal to the maximum potential energy at the extreme position.

Let us assume the motion to be simple Harmonic

$$x = A \sin \omega_n t$$

Differentiating w.r.t time, we get.

$$\dot{x} = \omega_n A \cos \omega_n t$$

To get the maximum velocity at the mean position, $\cos \omega_n t = 1$

$$\therefore \dot{x} = \omega_n A$$

$$\text{So Maximum Kinetic energy at mean position} = \frac{1}{2} m \dot{x}^2 \Rightarrow \frac{1}{2} m \omega_n^2 A^2 \rightarrow \textcircled{1}$$

$$\text{And Maximum potential energy at the extreme position} = \frac{1}{2} K A^2 \rightarrow \textcircled{2}$$

Equating $\textcircled{1}$ & $\textcircled{2}$, we get.

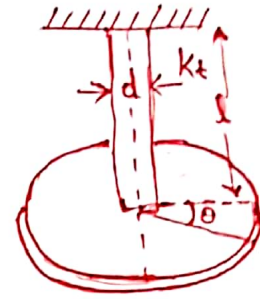
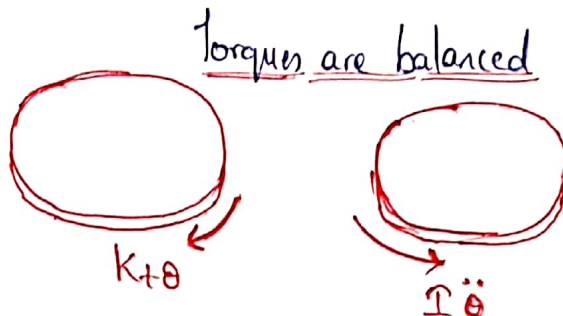
$$\frac{1}{2} m \omega_n^2 A^2 = \frac{1}{2} K A^2$$

$$\boxed{\omega_n = \sqrt{K/m}}$$

Torsional Vibrations

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Consider a system having a rotor of mass moment of Inertia (I) connected to a shaft of torsional stiffness ' k_t ' which is twisted by an angle ' θ ' as shown below



The body is rotated through an angle ' θ ' and released, the torsional vibration will result.

Note: Mass moment of Inertia of shaft is negligible when compared with that of rotor.

$$\therefore \ddot{\theta} + \frac{k_t}{I} \theta = 0 \quad \left[\because -k_t \theta = I \ddot{\theta} \right]$$

It can be taken as $\omega_n^2 = \frac{k_t}{I}$

$$\therefore \ddot{\theta} + \omega_n^2 \theta = 0$$

And natural frequency can be determined by $\omega_n = \sqrt{\frac{k_t}{I}}$
The torsional stiffness (k_t) of shaft can be determined with the help of the relation.

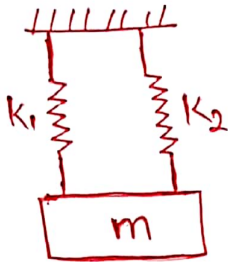
$$\frac{T}{J} = \frac{G\theta}{l} \quad \text{or} \quad k_t = \frac{T}{\theta} = \frac{GJ}{l}$$

$$k_t = \frac{GJ}{l} = \frac{G}{l} \cdot \frac{\pi}{32} \cdot d^4$$

Equivalent Stiffness of Spring Combinations

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In reality structural system can be considered as a system having springs. The springs can be joined parallel (or) in series. These no. of springs can be replaced by a single spring of the equivalent stiffness.

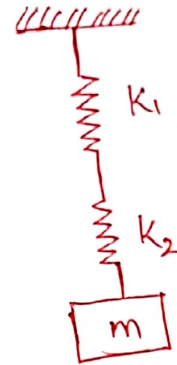


Springs in parallel

The deflection of individual spring is equal to the deflection of the system

$$\text{So } K_1 x + K_2 x = K_e x$$

$$\boxed{K_e = K_1 + K_2}$$



Springs in Series

The total deflection of the system is equal to the sum of deflection of individual springs

$$x = x_1 + x_2 + x_3 + \dots$$

$$\therefore \frac{F}{K_e} = \frac{F}{K_1} + \frac{F}{K_2} + \frac{F}{K_3} + \dots$$

$$\boxed{\frac{1}{K_e} = \frac{1}{K_1} + \frac{1}{K_2} + \dots}$$

Problems

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Q] Determine the natural frequency of the mass placed at one end of a cantilever beam of negligible mass as shown in figure

Solution: Stiffness of beam is given as

$$K = \frac{3EI}{l^3} \quad \text{where } EI \rightarrow \text{flexural rigidity}$$



The general equation of motion for undamped free vibrations is

$$m\ddot{x} + Kx = 0$$

Substituting for 'K', we have.

$$m\ddot{x} + Kx = 0 \quad \therefore m\ddot{x} + \frac{3EI}{l^3} x = 0$$

$$\omega_n = \sqrt{\frac{3EI}{l^3 m}} \quad \text{rad/sec}$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{3EI}{l^3}}$$

From Beam equation, we know

$$\frac{\partial^2 y}{\partial x^2} = \frac{M(x)}{EI}$$

$$\therefore EI \cdot \frac{\partial^2 y}{\partial x^2} = M = -P \cdot x$$

where $P \rightarrow$ transverse load.

Integrating we have

$$EI \cdot \frac{\partial y}{\partial x} = -\frac{Px^2}{2} + C_1$$

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~~@ $x=0$, $y=0$~~ ~~@ $x=l$, $y=?$~~

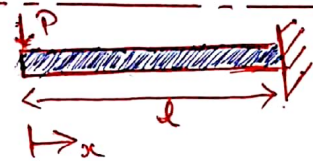
~~@ $x=0$, $\theta=0$~~ ~~@ $x=l$, $\theta=?$~~

~~$\therefore EI \cdot \theta(0) = -P$~~

@ $x=l$, $y=0$

@ $x=l$, $\theta=0 = \frac{\partial y}{\partial x}$

$$\therefore EI(0) = -\frac{Pl^2}{2} + C_1$$



$$\therefore C_1 = \frac{Pl^2}{2}$$

Integrating again w.r.t x ,

$$EI \cdot [y] = -\frac{Px^3}{6} + C_1 x + C_2$$

$$EI \cdot [0] = -\frac{Pl^3}{6} + \frac{Pl^3}{2} + C_2$$

$$\therefore C_2 = -\frac{Pl^3}{3}$$

Substituting in main equation, we have.

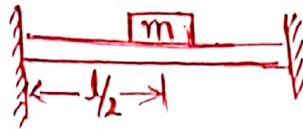
$$EI(y) = -\frac{Pl^3}{6} + \frac{Pl^3}{2} = \frac{Pl^3}{3}$$

$$\therefore @ x=0$$

$$y = -\frac{Pl^3}{3EI}$$

from above equation $K = \frac{P}{y} = \frac{3EI}{l^3} \checkmark$

[2] Find the natural frequency of the system shown in figure below



[Pg-8]

Solution:

$$\text{deflection} = \frac{Fl^3}{192EI}$$

The general equation for undamped free vibration is
 $m\ddot{x} + Kx = 0$

$$\omega_n = \sqrt{K/m} = \sqrt{\frac{192EI}{ml^3}} \text{ rad/sec}$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{192EI}{ml^3}}$$

[3] Determine the effect of the mass of the spring on the natural frequency of the system shown.

Solution:

Let $x \rightarrow$ displacement

$m \rightarrow$ Mass

$\therefore \dot{x} \rightarrow$ velocity of spring element at distance 'y' from fixed end is written as $\left(\frac{\dot{x}y}{l}\right)$

$\therefore$ Kinetic energy of spring element dy is written as
 $\frac{1}{2} m (\dot{x})^2 \Rightarrow \frac{1}{2} (\delta \cdot dy) \cdot \left(\frac{\dot{x}y}{l}\right)^2$

$\delta \rightarrow$ Mass of spring per unit length

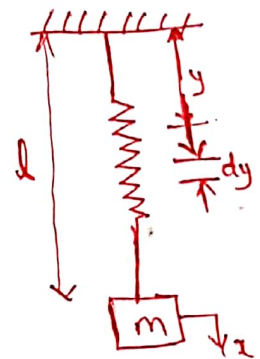
$\therefore$ Total Kinetic energy is written as.

$$KE = \frac{1}{2} m \dot{x}^2 + \int_0^l \frac{1}{2} (\delta \cdot dy) \left(\frac{y}{l} \dot{x}\right)^2$$

$$= \frac{1}{2} m \dot{x}^2 + \int_0^l \frac{1}{2} \delta \frac{y^2}{l^2} \cdot \dot{x}^2 \cdot dy$$

$$= \frac{1}{2} m \dot{x}^2 + \frac{1}{2} \frac{\delta}{l^2} \cdot \dot{x}^2 \cdot \left[\frac{y^3}{3}\right]_0^l \Rightarrow \frac{1}{2} m \dot{x}^2 + \frac{1}{2} \delta \dot{x}^2 \frac{l}{3}$$

Let us take mass of spring $m_s = \delta l$



$$\delta = \frac{m}{l}$$

$$\delta = \frac{m}{dy}$$

$$\therefore m = \delta \cdot l$$

$$\therefore KE = \frac{1}{2} m \dot{x}^2 + \frac{1}{6} m_0 \dot{x}^2$$

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$$\text{Potential Energy of the system} = \frac{1}{2} K x^2$$

$$\therefore \text{Total Energy} = KE + PE$$

$$\text{ie } \frac{1}{2} m \dot{x}^2 + \frac{1}{6} m_0 \dot{x}^2 + \frac{1}{2} K x^2 = \text{Constant.}$$

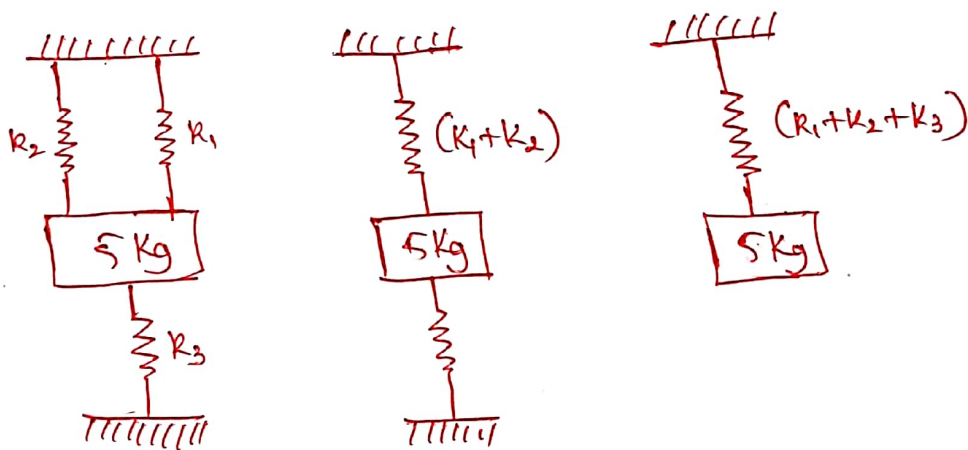
Differentiating the above, w.r.t time

$$m \ddot{x} + \frac{m_0 \ddot{x}}{3} + K x = 0$$

$$\left(m + \frac{m_0}{3}\right) \ddot{x} + K x = 0$$

$$\omega_n = \sqrt{\frac{K}{\left(m + \frac{m_0}{3}\right)}} \text{ rad/sec.}$$

- ② Find the natural frequency of the system shown in figure
Given $K_1 = K_2 = 1500 \text{ N/m}$, $K_3 = 2000 \text{ N/m}$, $m = 5 \text{ Kg}$



Solution

$$K_e = K_1 + K_2 + K_3$$

$$= 1500 + 1500 + 2000 \Rightarrow 5000 \text{ N/m}$$

$$\omega_n = \sqrt{\frac{K}{m}} = \sqrt{\frac{5000}{5}} = 31.62 \text{ rad/sec}$$

$$f = 5.03 \text{ Hz}$$

- ⑤ An unknown mass 'm' is attached to one end of a spring of stiffness 'k' having natural frequency of 6 Hz. When 1 kg mass is attached with 'm' the natural frequency of the system is lowered by 20%. Determine the value of unknown mass 'm' and stiffness 'k'. Pg-10

Solution:

$$f_1 = 6 \text{ Hz} = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$f_2 = 4.8 \text{ Hz} = \frac{1}{2\pi} \sqrt{\frac{k}{m+1}}$$

$$20\% = 1.2$$

$$\therefore 20\% \text{ reduction} = 6 - 1.2 = 4.8$$

$$\therefore \frac{f_1}{f_2} = \frac{6}{4.8} = \frac{\sqrt{k/m}}{\sqrt{k/(m+1)}}$$

$$(1.25)^2 = \frac{k/m}{(k/(m+1))} \Rightarrow \frac{m+1}{m}$$

$$\therefore m = 1.777 \text{ kg}$$

Stiffness from equ (1) can be determined.

$$6 = \frac{1}{2\pi} \sqrt{\frac{k}{1.777}}$$

$$\therefore k = 2523.94 \text{ N/m}$$

- ⑥ A spring-mass system ($k_1 - m_1$) has a natural frequency f_1 . Calculate the value of k_2 another spring which when connected to k_1 in parallel increases the frequency by 30%.

Solution:

$$f_1 = \sqrt{k_1/m_1}$$

If another spring is connected to k_1 in parallel, then $k_e = k_1 + k_2$

$$f_2 = \frac{130}{100} f_1 = \sqrt{\frac{k_1 + k_2}{m_1}}$$

$$\frac{f_1}{f_2} = \frac{1}{1.3} = \sqrt{\frac{k_1}{m_1} \cdot \frac{m_1}{k_1 + k_2}}$$

$$\frac{1}{1.3} = \sqrt{\frac{k_1}{k_1 + k_2}}$$

$$1.69 k_1 = k_1 + k_2$$

$$k_2 = 0.69 k_1$$

[7] An electric motor is supported by six springs of stiffness 'k' each. The moment of Inertia of the motor is 'I'. Determine the natural frequency of the system.

Solution:

Restoring torque = Spring force
× displacement

$$\tau = [6kb\theta] \times b$$

∵ Springs in parallel i.e. $K_1x + K_2x = Kx$

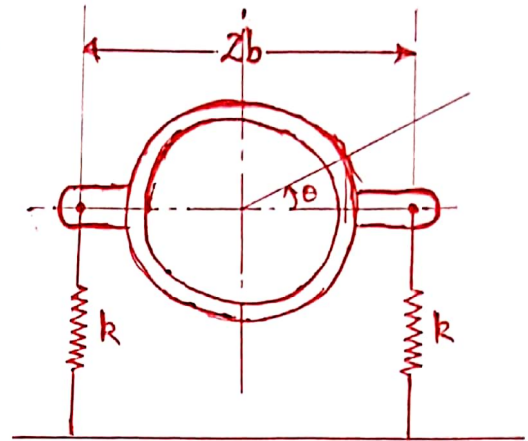
$$\text{i.e. } [k+k+k+k+k+k] \times \text{displacement} = 6K(\theta b)$$

$$\therefore \text{displ} = \theta \times r$$

$$\therefore I \ddot{\theta} = -6kb^2\theta$$

$$\ddot{\theta} + \frac{6kb^2}{I}\theta = 0$$

$$\therefore \omega_n = \sqrt{\frac{6kb^2}{I}} = b \sqrt{\frac{6k}{I}} \text{ rad/sec.}$$



[8] Determine the natural frequency of the spring mass pulley system shown in figure.

Solution:

$$\text{Total Energy (T)} = \text{KE of mass (m)} + \text{KE of pulley (M)} + \text{PE of spring k}$$

From free body diagram, it is clear that at any moment $x = r\theta$

$$\text{Moment of inertia of pulley } I = \frac{1}{2} Mr^2$$

Differentiating Total Energy equation w.r.t time, we get

where $\cancel{P = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} kx^2}$

$$P = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} kx^2$$

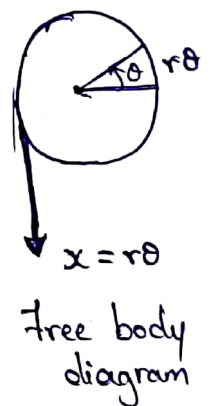
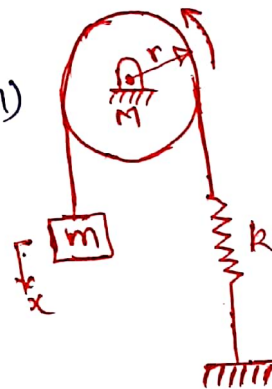
$$P = \frac{1}{2} m r^2 \dot{\theta}^2 + \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} k r^2 \theta^2 = \text{constant}$$

differentiate

$$m r^2 \ddot{\theta} + I \ddot{\theta} + k r^2 \theta = 0$$

$$m r^2 \ddot{\theta} + I \ddot{\theta} + k r^2 \theta = 0$$

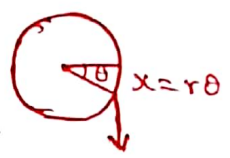
$$(m r^2 + \frac{1}{2} M r^2) \ddot{\theta} + k r^2 \theta = 0$$



$$\therefore \theta + \frac{kr^2}{mr^2 + \frac{1}{2}Mr^2} \theta = 0$$

$$\omega_n = \sqrt{\frac{2Kr^2}{2mr^2 + Mr^2}} = \sqrt{\frac{2k}{2m+M}} \text{ rad/sec}$$

[9] A circular cylinder of mass 4 kg and radius 15 cm is connected by a spring of stiffness 4000 N/m as shown in figure. It is free to roll on horizontal rough surface without slipping, determine the natural frequency.



Total Energy = KE due to Translatory motion + KE due to rotary motion + PE of spring

$$= \frac{1}{2} m \dot{x}^2 + \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} kx^2$$

$$= \frac{1}{2} mr^2 \dot{\theta}^2 + \frac{1}{2} \cdot \frac{1}{2} mr^2 \dot{\theta}^2 + \frac{1}{2} Kr^2 \theta^2$$

$$T = \frac{3}{4} mr^2 \dot{\theta}^2 + \frac{1}{2} Kr^2 \theta^2 = \text{Constant}$$

Differentiating w.r.t time we get.

$$\frac{3}{4} \cdot 2 mr^2 \dot{\theta} \ddot{\theta} + Kr^2 \dot{\theta} \theta = 0$$

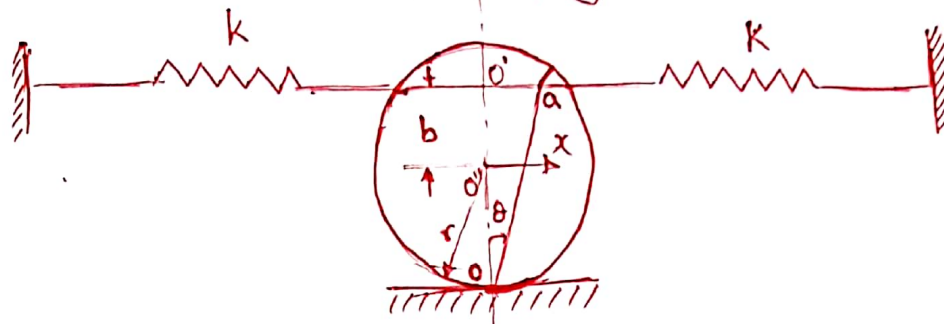
$$\frac{3}{2} mr^2 \ddot{\theta} + Kr^2 \theta = 0$$

$$\omega_n = \sqrt{\frac{Kr^2}{3/2 mr^2}} = \sqrt{\frac{2k}{3m}} \text{ rad/sec.}$$

$$f_n = \frac{1}{2\pi} \sqrt{\frac{2 \times 4000}{3 \times 4}}$$

$$f_n = 4.11 \text{ Hz}$$

[10] Find the natural frequency of the system shown in figure.



$$\begin{aligned}
 KE &= \frac{1}{2} m r^2 \dot{\theta}^2 + \frac{1}{2} I \dot{\theta}^2 \\
 &= \frac{1}{2} m r^2 \dot{\theta}^2 + \frac{1}{4} m r^2 \dot{\theta}^2 \\
 &\Rightarrow \frac{3}{4} m r^2 \dot{\theta}^2
 \end{aligned}$$

$$\text{Moment of Inertia (I)} = \frac{1}{2} m r^2$$

From geometry of figure.

$$x = O'a \Rightarrow O'O \times \theta \Rightarrow (O'O'' + O'O') \times \theta$$

$$x = (r+b)\theta$$

$$PE = \frac{1}{2} k x^2 + \frac{1}{2} k x^2$$

$$= 2 \cdot \frac{1}{2} k [(r+b)\theta]^2 = k(r+b)^2 \theta^2$$

$$\therefore KE + PE = \text{Constant}$$

$$\frac{3}{4} m r^2 \dot{\theta}^2 + k(r+b)^2 \theta^2 = \text{Constant}$$

$$\therefore 2 \times \frac{3}{4} m r^2 \ddot{\theta} + k(r+b)^2 2\theta = 0$$

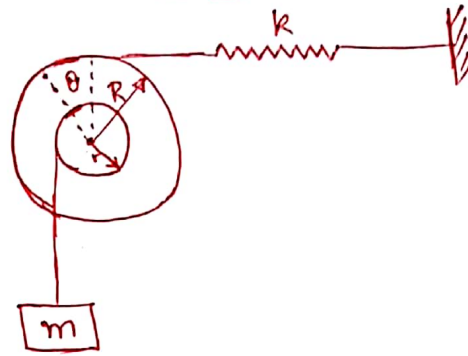
$$\therefore \frac{3}{4} m r^2 \ddot{\theta} + k(r+b)^2 \theta = 0$$

Hence the frequency of vibration is.

$$\frac{1}{2\pi} \sqrt{\frac{4k(r+b)^2}{3mr^2}} \text{ Hz}$$

Find the natural frequency of the system shown in figure.

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Solution: Let $x_2 \rightarrow$ deflection of spring $= R \cdot \theta$

$$\text{Spring force} = kx_2 = kR \cdot \theta.$$

$$x_1 = r \cdot \theta$$

$$\begin{aligned} \text{Total Kinetic Energy} &= \text{KE of the mass} + \text{KE of rotating element.} \\ &= \frac{1}{2} m \dot{x}_1^2 + \frac{1}{2} I \dot{\theta}^2 \end{aligned}$$

$$\text{Potential Energy of spring} = \frac{1}{2} kx_2^2$$

$$\text{Total Energy} = \frac{1}{2} m \dot{x}_1^2 + \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} kx_2^2$$

from energy method, we get.

$$\frac{1}{2} m \dot{x}_1^2 + \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} kx_2^2 = \text{Constant.}$$

$$\text{ie } \frac{d}{d\theta} \left[\frac{1}{2} m r^2 \dot{\theta}^2 + \frac{1}{2} I \dot{\theta}^2 + \frac{1}{2} k R^2 \theta^2 \right] = \frac{d}{d\theta} (\text{const.})$$

$$m r^2 \ddot{\theta} + I \ddot{\theta} + k R^2 \theta = 0$$

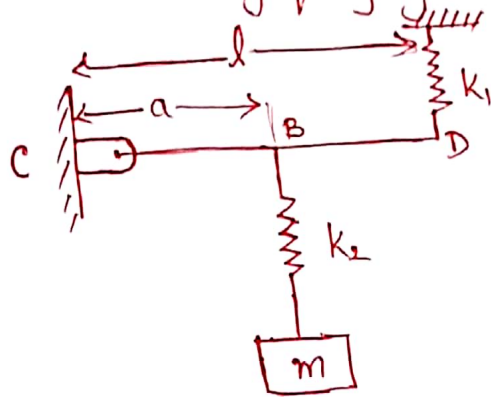
$$\text{ie } (m r^2 + I) \ddot{\theta} + k R^2 \theta = 0$$

$$\ddot{\theta} + \left(\frac{k R^2}{m r^2 + I} \right) \theta = 0$$

$$\omega_n = \sqrt{\frac{k R^2}{m r^2 + I}}$$

[12] Find the natural frequency of the system in figure

[Assuming the bar CD to be weightless and rigid]



Let us consider a force P at D, so the force at point B can be written as $\frac{Pl}{a}$

$$F_B \cdot a = P \cdot l$$

$$F_B =$$

Damping: The resistance offered by a body against vibration is called as damping. The resistance may be internally (or) externally applied, simultaneously it can be either liquid (or) solid (or) Combination of both.

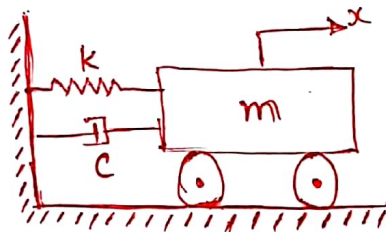
There are mainly four types of damping used in mechanical system.

- i) Viscous damping
- ii) Coulomb damping
- iii) Structural damping
- iv) Non-linear, Slip (or) interfacial damping

Differential equations of damped free vibration

The best example to be considered to do vibration analysis is a mechanical system having spring, mass and damper system as shown in figure below.

It is a well known fact that viscous force is proportional to the velocity across the damper and hence it is written as $c \cdot \frac{dx}{dt} = c\dot{x}$



Free body diagram

Equation of motion is written as

$$m\ddot{x} + c\dot{x} + Kx = 0$$

→ Characteristic equation

Assuming a solution $x = e^{ut}$

where $u \rightarrow$ Constant to be determined.

$$\therefore \dot{x} = ue^{ut} \text{ and } \ddot{x} = u^2 e^{ut}$$

$$\therefore \boxed{mu^2 e^{ut} + cue^{ut} + ke^{ut} = 0}$$

$$u^2 + \frac{cu}{m} + \frac{K}{m} = 0$$

Solving the above equation for 'u'

$$u = \frac{-c/m \pm \sqrt{c^2/m^2 - 4k/m}}{2}$$

$$\therefore x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\therefore u = \frac{-c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 - k/m}$$

$\therefore$ the two roots are

$$u_1 = \frac{-c}{2m} + \sqrt{\left(\frac{c}{2m}\right)^2 - k/m} \quad \text{and}$$

$$u_2 = \frac{-c}{2m} - \sqrt{\left(\frac{c}{2m}\right)^2 - k/m}$$

The solution of equation is written as

$$x = A_1 e^{u_1 t} + A_2 e^{u_2 t}$$

where A_1 and A_2 are arbitrary constants

$$\therefore x = A_1 e^{[-c/2m + \sqrt{(c/2m)^2 - k/m}]t} + A_2 e^{[-c/2m - \sqrt{(c/2m)^2 - k/m}]t}$$

Critical damping Constant and Damping ratio

From the above equation, it is clear that only $(c/2m)^2 - k/m$ is constant.

Hence critical damping (c_c) is defined as the value of damping Co-efficient ' c ' for which $(\frac{c}{2m})^2 - k/m$ is equal to zero.

$$\therefore \left(\frac{c_c}{2m}\right)^2 = \frac{k}{m} \quad \text{ie} \quad \frac{c_c}{2m} = \sqrt{\frac{k}{m}}$$

$$\therefore c_c = 2m \sqrt{k/m} \Rightarrow 2m\omega$$

The ratio ' c ' to ' c_c ' is termed as damping ratio (ϵ).

$$\text{ie} \quad \epsilon = \frac{c}{c_c}$$

$$\frac{c}{2m} \text{ can be also written as } \left(\frac{c}{c_c}\right)\left(\frac{c_c}{2m}\right) = \epsilon\omega$$

$$\therefore x_1 = A_1 e^{(-\epsilon + \sqrt{\epsilon^2 - 1})\omega t} + A_2 e^{(-\epsilon - \sqrt{\epsilon^2 - 1})\omega t}$$

Based on the value of damping ratio ϵ , the damped systems are put into three categories

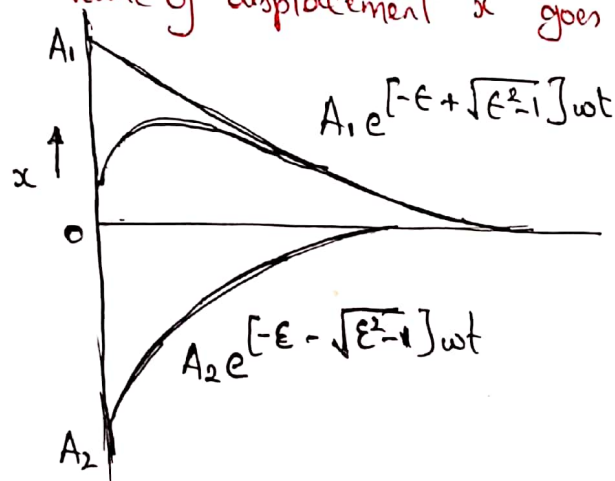
i) Over damped system ($\epsilon > 1$)

When the value of damping ratio is greater than '1', the system is known as over-damped ~~an~~ system.

→ The motion is aperiodic

→ When $t=0$, the displacement is the sum of A_1 and A_2
ie $x = A_1 + A_2$

→ The value of displacement x goes on decreasing with time.



→ The system is non-vibratory in nature

→ The system comes to equilibrium in an exponential manner.

→ Once the system is disturbed, it will take infinite time to come back to equilibrium position.

→ The values of A_1 and A_2 can be determined from initial conditions.

ii) Critically Damped System

The system is said to be critically damped when $\epsilon = 1$ ie

$$\frac{c}{2m} = \sqrt{\frac{k}{m}}$$

The roots of the equation u_1 and u_2 are equal

$$\text{ie } u_1 = u_2 = -\epsilon\omega = -\omega$$

So the approximate solution is written as

$$x = A_1 e^{-\omega t} + A_2 t e^{-\omega t} \Rightarrow (A_1 + A_2 t) e^{-\omega t}$$

Applying initial condition to above equation

$$x_0 = A_1$$

$$\dot{x}_0 = \left(\frac{dx}{dt}\right) = A_1(-\omega) + A_2$$

$$\dot{x}_0 = -\omega A_1 + A_2$$

$$\dot{x}_0 = -\omega x_0 + A_2$$

$$A_2 = \dot{x}_0 + \omega x_0$$

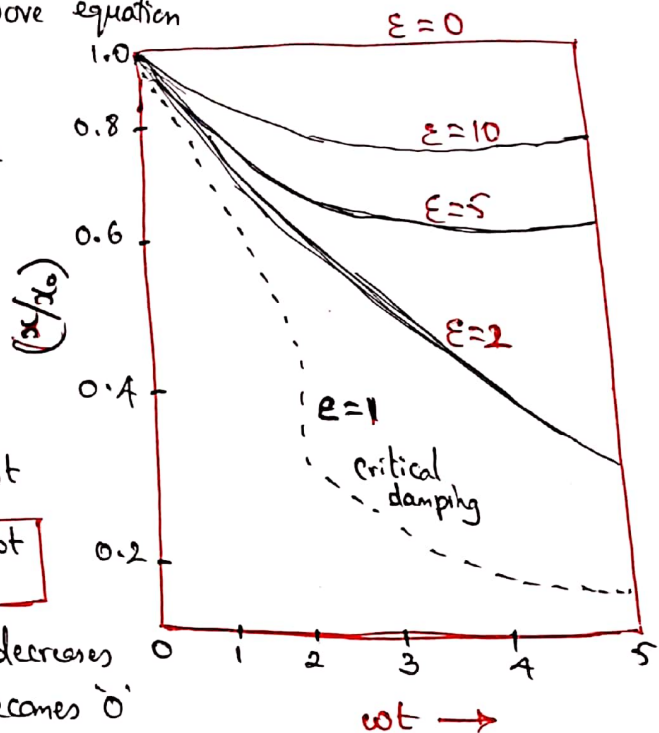
So solution becomes

$$x = x_0 e^{-\omega t} + (\dot{x}_0 + \omega x_0) t e^{-\omega t}$$

$$x = [x_0 + (\dot{x} + \omega x_0)t] e^{-\omega t}$$

We see that the value of $\dot{x}$ decreases as t increases and finally becomes 0 as t tends to infinity.

→ This is also an aperiodic motion



iii) Under damped System

In underdamped system the value of damping ratio ' ξ ' is less than one. Therefore the roots of the equation can be written as

$$u_1 = [-\xi + j\sqrt{1-\xi^2}]\omega$$

$$u_2 = [-\xi - j\sqrt{1-\xi^2}]\omega$$

where $j = \sqrt{-1} \rightarrow$ Imaginary unit of Complex roots

$\therefore$ The displacement can be written as

$$x = A_1 e^{[-\xi + j\sqrt{1-\xi^2}]\omega t} + A_2 e^{[-\xi - j\sqrt{1-\xi^2}]\omega t}$$

~~$$x = e^{-\xi\omega t}$$~~

$$x = e^{-\xi\omega t} [A_1 e^{j\sqrt{1-\xi^2}\omega t} + A_2 e^{-j\sqrt{1-\xi^2}\omega t}]$$

we know that $e^{jx} = \cos x + j \sin x$

$$\therefore x = e^{-\xi\omega t} [A_1 \cos \sqrt{1-\xi^2}\omega t + A_1 j \sin \sqrt{1-\xi^2}\omega t + A_2 \cos \sqrt{1-\xi^2}\omega t - A_2 j \sin \sqrt{1-\xi^2}\omega t]$$

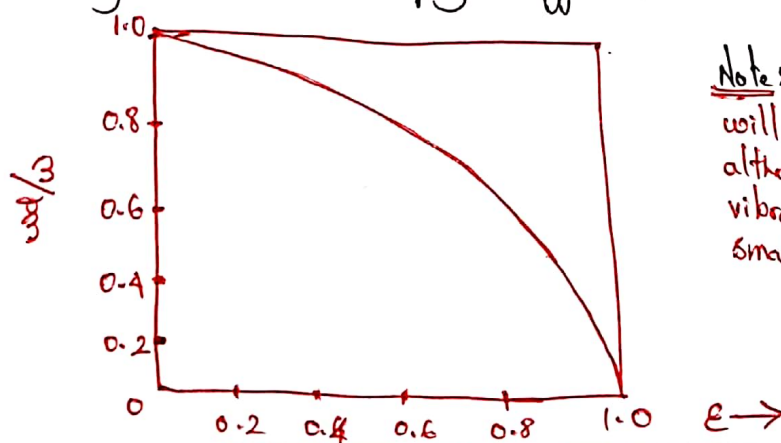
$$x = e^{-\xi\omega t} [C_1 \cos \sqrt{1-\xi^2}\omega t + C_2 \sin \sqrt{1-\xi^2}\omega t]$$

where $C_1 = A_1 + A_2$ and $C_2 = (A_1 - A_2)j$

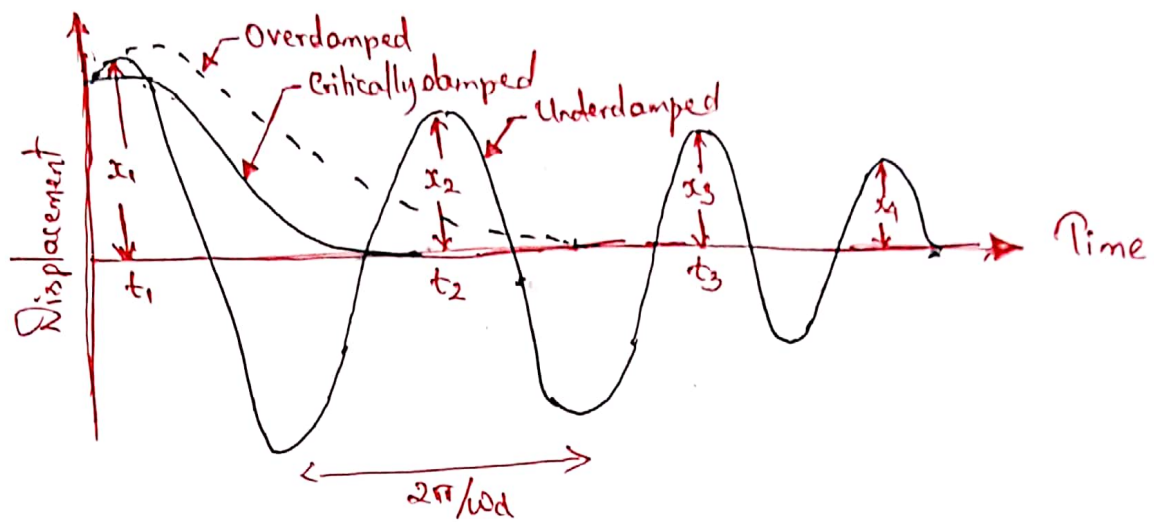
and $x = C_3 e^{-\xi\omega t} \sin(\sqrt{1-\xi^2}\omega t + \phi_1)$ or
 $x = C_4 e^{-\xi\omega t} \cos(\sqrt{1-\xi^2}\omega t + \phi_2)$

$\rightarrow$ From the equation, it is clear that, The amplitude decreases exponentially with time

The term $\sqrt{1-\xi^2} \cdot \omega = \omega_d =$ Damped natural frequency
 The variation of ω_d with damping Co-efficient ξ is shown below



Note: Theoretically the S/m will never come to rest although the amplitude of vibration may be very very small.



$$\frac{2\pi}{\omega\sqrt{1-\xi^2}} \quad \text{or} \quad \frac{2\pi}{\omega_d} = T_d \rightarrow \text{Time period. - damped}$$

Logarithmic Decrement

It is defined as the natural logarithm of the ratio of any two successive amplitudes on the same side of the mean line.

As per the definition from above figure, logarithmic decrement (δ) is written as

$$\delta = \ln\left(\frac{x_1}{x_2}\right)$$

$$\text{Taking } x = C_1 e^{-\xi\omega t} \cos(\sqrt{1-\xi^2} \cdot \omega t + \phi_2)$$

where t_1 and t_2 denote the times corresponding to two successive amplitudes.

$$\therefore \frac{x_1}{x_2} = \frac{C_1 e^{-\xi\omega t_1} \cos(\sqrt{1-\xi^2} \omega t_1 + \phi_2)}{C_1 e^{-\xi\omega t_2} \cos(\sqrt{1-\xi^2} \omega t_2 + \phi_2)}$$

$$\frac{x_1}{x_2} = e^{-\xi\omega(t_1 - t_2)} \cdot \frac{\cos(\sqrt{1-\xi^2} \omega t_1 + \phi_2)}{\cos(\sqrt{1-\xi^2} \omega t_2 + \phi_2)}$$

Let us assume $t_2 = t_1 + T_d$

$$\text{where } T_d = \frac{2\pi}{\omega_d}$$

$$\therefore \frac{\cos(\omega_d t_1 + \phi_2)}{\cos(\omega_d (t_1 + T_d) + \phi_2)} = \frac{\cos(\omega_d t_1 + \phi_2)}{\cos\left\{\omega_d \left(t_1 + \frac{2\pi}{\omega_d}\right) + \phi_2\right\}}$$

$$\therefore \Rightarrow \frac{\cos(\omega_d t_1 + \phi_2)}{\cos\{(\omega_d t_1 + 2\pi) + \phi_2\}} = 1$$

$$\therefore \frac{x_1}{x_2} = e^{-\epsilon \omega (t_1 - t_1 - t_d)} \Rightarrow e^{\epsilon \omega t_d} \Rightarrow e^{\frac{\epsilon \omega 2\pi}{\omega_d}}$$

$$\frac{x_1}{x_2} = e^{\frac{\epsilon \omega 2\pi}{\sqrt{1-\epsilon^2} \cdot \omega}} \Rightarrow e^{2\pi \epsilon / \sqrt{1-\epsilon^2}}$$

$$\therefore \delta = \log_e \left(\frac{x_1}{x_2} \right) = \frac{2\pi \epsilon}{\sqrt{1-\epsilon^2}}$$

When the value of ' ϵ ' is very small, the above equation is written as

$$\delta = 2\pi \epsilon$$

If the system executes 'n' cycles, the logarithmic decrement ' δ ' is written as

$$\delta = \frac{1}{n} \log_e \frac{x_1}{x_{n+1}}$$

$x_1 \rightarrow$ Amplitude at starting point
 $x_{n+1} \rightarrow$ Amplitude after 'n' Cycles.



$$\frac{x_1}{x_2} = \frac{x_2}{x_3} = \frac{x_3}{x_4} = \dots = \frac{x_{n-1}}{x_n} = e^{\delta}$$

$$\left(\frac{x_1}{x_2} \right) \left(\frac{x_2}{x_3} \right) \left(\frac{x_3}{x_4} \right) \dots \left(\frac{x_n}{x_{n+1}} \right) = (e^{\delta})^n$$

$$\therefore n\delta = \ln \frac{x_1}{x_{n+1}}$$

$$\delta = \frac{1}{n} \ln \frac{x_1}{x_{n+1}}$$

Vibrational Energy and Logarithmic decrement

Let us consider at any given instant x_1 be the amplitude and after one cycle, it will be x_2

$$\therefore \delta = \ln \frac{x_1}{x_2}$$

$$\frac{x_1}{x_2} = e^{\delta}$$

$$\frac{x_2}{x_1} = e^{-\delta}$$

On Expanding, we get

$$e^{-\delta} = 1 - \delta + \frac{\delta^2}{2!} - \frac{\delta^3}{3!} + \dots$$

Let E_1 be the vibrational energy at amplitude x_1

$$E_1 = \frac{1}{2} K x_1^2$$

Similarly $E_2 \rightarrow$ Vibrational energy at amplitude x_2 , so

$$E_2 = \frac{1}{2} K x_2^2$$

$$\text{Now } \frac{E_1 - E_2}{E_1} = 1 - \frac{E_2}{E_1} \Rightarrow 1 - \frac{x_2^2}{x_1^2} \Rightarrow 1 - e^{-2\delta}$$

$$= 1 - \left[1 - 2\delta + \frac{(2\delta)^2}{2!} - \frac{(2\delta)^3}{3!} + \dots \right]$$

$$= 2\delta - \frac{(2\delta)^2}{2!} + \frac{(2\delta)^3}{3!}$$

For small δ , higher powers of δ may be neglected.

$$\text{So } \frac{E_1 - E_2}{E_1} = 2\delta$$

$$\boxed{\delta = \frac{\Delta E}{2E_1}}$$

where $\Delta E = E_1 - E_2 =$ Energy dissipated in one cycle.

Unit - II - Problems

- ① A damping force having magnitude $2 \cos(2\pi t - \pi/4) \text{ N}$, gives $5 \cos 2\pi t \text{ m}$ displacement. Calculate

i) The energy dissipated during first 5-seconds

ii) Energy dissipated during the first $3/4 \text{ sec}$.

Solution: From the given data, it is clear that, the displacement leads the force by $\pi/4 = 45^\circ$.

$$F = F_0 \cos(\omega t - \phi) = 2 \cos(2\pi t - \pi/4)$$

$$x = A \cos \omega t = 5 \cos 2\pi t$$

$$\therefore F_0 = 2, \omega = 2\pi \text{ and } \phi = \pi/4$$

$$A = 5$$

Time period of force 'F' and displacement 'x' is

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2\pi} = 1 \text{ sec}$$

i) In 5-seconds, there will be 5-Complete Cycles, therefore.

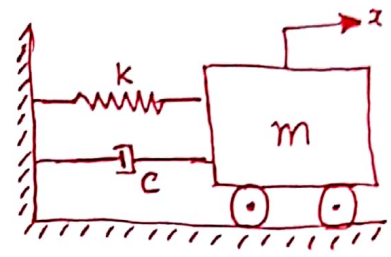
$$\text{Energy dissipated} = n(\pi) F_0 A \sin \frac{\pi}{4}$$

$$= 5(\pi \times 2 \times 5 \times \sin 45^\circ) = \underline{\underline{111.01 \text{ N-m}}}$$

ii) During $3/4 \text{ sec}$ only part of cycle is complete. To find the energy during this period, we have

$$\begin{aligned} \text{Energy dissipation} &= \int_0^t F \cdot \frac{dx}{dt} \cdot dt = \int_0^{3/4} \left[2 \cos\left(2\pi t - \frac{\pi}{4}\right) \frac{d}{dt} 5 \cos 2\pi t \right] dt \\ &\Rightarrow \int_0^{3/4} \left[2 \cos\left(2\pi t - \frac{\pi}{4}\right) (-5 \times 2\pi) \sin 2\pi t \right] dt \Rightarrow \int_0^{3/4} (-20\pi) \cos\left(2\pi t - \frac{\pi}{4}\right) \sin 2\pi t dt \\ &\Rightarrow \int_0^{3/4} (-20\pi) \left[\cos 2\pi t \cdot \cos \frac{\pi}{4} + \sin 2\pi t \sin \frac{\pi}{4} \right] (\sin 2\pi t) dt \\ &\Rightarrow \int_0^{3/4} (-44.406) \left[\frac{1}{2} \sin 4\pi t + \frac{2}{2} \sin^2 2\pi t \right] dt = \underline{\underline{-40.62 \text{ N-m}}} \end{aligned}$$

- ② Find the equation of motion for the system shown in figure, when
 (a) $\varepsilon = 1.0$, (b) $\varepsilon = 0.3$ and (c) $\varepsilon = 2.0$, if the mass m is displaced
 by a distance of 3 cm and released.



a) When $\varepsilon = 1.0$ [Critically damped System]

$$x = (A_1 + A_2 t) e^{-\omega t}$$

Given displacement $\bar{x} = 3 \text{ cm}$ at $t = 0$,

$$\therefore 3 = A_1$$

Differentiating eqn (1) w.r.t $\bar{x}$, we get

$$\dot{x} = -\omega A_1 e^{-\omega t} + A_2 e^{-\omega t} - \omega A_2 t e^{-\omega t}$$

@ $t = 0$, we have $\dot{x} = 0$

$$0 = -\omega A_1 + A_2$$

$$\therefore A_2 = +\omega A_1 = 3\omega$$

$\therefore$ therefore displacement $\bar{x}$ is given as.

$$x = (3 + 3\omega t) e^{-\omega t} = 3(1 + \omega t) e^{-\omega t}$$

b) When $\varepsilon = 0.3$ [Under damped S/m]

$$x = A_1 e^{-\varepsilon \omega t} \sin(\sqrt{1 - \varepsilon^2} \omega t + \phi) \Rightarrow A_1 e^{-0.3 \omega t} \sin(\sqrt{1 - 0.09} \omega t + \phi)$$

$$x = A_1 e^{-0.3 \omega t} \sin(0.953 \omega t + \phi)$$

@ $t = 0$, $x = 3$

$$\therefore 3 = A_1 \sin \phi$$

Differentiating the displacement equation, we get.

$$\dot{x} = A_1 [-0.3 \omega e^{-0.3 \omega t} \sin(0.953 \omega t + \phi) + e^{-0.3 \omega t} (0.953) \omega \cos(0.953 \omega t + \phi)]$$

@ $t = 0$, $\dot{x} = 0$

$$\dot{x} = -0.3 \omega A_1 e^{-0.3 \omega t} \sin(0.953 \omega t + \phi) + A_1 e^{-0.3 \omega t} (0.953) \omega \cos(0.953 \omega t + \phi)$$

@ $t = 0$, $\dot{x} = 0$

$$\therefore 0 = -0.3 \omega A_1 \sin(\phi) + A_1 (0.953 \omega) \cos(\phi)$$

$$\therefore A_1 \cos(\phi) \times 0.953 \cancel{\omega} = A_1 \sin(\phi) \times 0.3 \cancel{\omega}$$

$$\tan(\phi) = \frac{0.953}{0.3} = 3.1766$$

$$\therefore \phi = \tan^{-1} 3.1766 = 72^\circ$$

$$\therefore A_1 = 3 / \sin \phi = 3 / \sin(72^\circ) = 3.145$$

$\therefore$ The equation of motion can be written as

$$x = 3.145 e^{-0.3\omega t} \sin(0.953\omega t + 72^\circ)$$

(c) When $\varepsilon = 2.0$ (Over damped g/m)

$$x = A_1 e^{[-\varepsilon + \sqrt{\varepsilon^2 - 1}]\omega t} + A_2 e^{[-\varepsilon - \sqrt{\varepsilon^2 - 1}]\omega t}$$

$$= A_1 e^{(-2 + \sqrt{3})\omega t} + A_2 e^{(-2 - \sqrt{3})\omega t}$$

Applying The initial Conditions, we get

@ $t=0$ ~~$x=0$~~ $x=3 = A_1 + A_2$

ie $A_1 + A_2 = 3$

Differentiating the displacement equation, we get .

~~$$\dot{x} = [-\varepsilon + \sqrt{\varepsilon^2 - 1}]\omega A_1 e^{[-\varepsilon + \sqrt{\varepsilon^2 - 1}]\omega t} + [-\varepsilon - \sqrt{\varepsilon^2 - 1}]\omega A_2 e^{[-\varepsilon - \sqrt{\varepsilon^2 - 1}]\omega t}$$~~

$$\dot{x} = (-2 + \sqrt{3})\omega A_1 e^{(-2 + \sqrt{3})\omega t} + (-2 - \sqrt{3})\omega A_2 e^{(-2 - \sqrt{3})\omega t}$$

@ $t=0$, $\dot{x}=0$

$$0 = (-2 + \sqrt{3})\omega A_1 + (-2 - \sqrt{3})\omega A_2$$

$$(2 + \sqrt{3})\omega A_2 = (-2 + \sqrt{3})\omega A_1$$

$$A_1 = -13.92 A_2$$

Substituting, we get $A_2 = -0.232$ and $A_1 = 3.232$

$$\therefore x = 3.232 e^{-0.2679\omega t} - 0.232 e^{-3.7322\omega t}$$

- ③ A damper offers resistance 0.05 N at constant velocity 0.04 m/s . The damper is used with $K = 9 \text{ N/m}$. Determine the damping and frequency of the system when the mass of the system is 0.10 kg .

Solution: Damping force $F = c\dot{x}$

$$\text{ie } 0.05 = c \times (0.04)$$

$$\therefore c = 1.25 \text{ N-sec/m}$$

$$\therefore \text{Critical damping} = c_c = 2\sqrt{Km} = 2\sqrt{9 \times 0.1}$$

$$c_c = 1.897 \text{ N-sec/m}$$

$$\therefore \text{Damping ratio} = \xi = \frac{c}{c_c} = 0.658$$

The system is underdamped system. The frequency of damped vibration is

$$\omega_d = \sqrt{1 - \xi^2} \cdot \omega = \sqrt{1 - (0.658)^2} \cdot \frac{9}{0.10}$$

$$\boxed{\omega_d = 7.14 \text{ rad/sec}}$$

- ④ A vibratory system in a vehicle is to be designed with the following parameters. $K = 100 \text{ N/m}$, $C = 2 \text{ N-sec/m}$, $m = 1 \text{ kg}$.

Calculate the (i) decrease of amplitude from its starting value after 3-Complete oscillations and (ii) the frequency of oscillations.

$$\omega = \sqrt{\frac{K}{m}} = 10 \text{ rad/sec} \quad \text{and} \quad c_c = 2\sqrt{Km} = 20 \text{ N-sec/m}$$

$$\xi = \frac{c}{c_c} = \frac{2}{20} = 0.10$$

$$\delta = \frac{2.5\xi}{\sqrt{1-\xi^2}} = \frac{2.5 \times 0.10}{\sqrt{1-0.01}} = 0.63$$

$$\delta = \ln \frac{x_1}{x_2} \quad \text{and for } n\text{-oscillation} \quad \delta = \frac{1}{n} \ln \frac{x_1}{x_{n+1}}, \text{ when } n=3$$

$$\delta = \frac{1}{3} \ln \frac{x_1}{x_{n+1}}$$

$$\therefore \frac{x_1}{x_{n+1}} = e^{(3 \times 0.63)} = e^{1.89} = 6.62 \quad \text{and} \quad \omega_d = \omega \sqrt{1-\xi^2} = 9.95 \text{ rad/sec}$$

⑤ A horizontal spring mass system with coulomb damping has a mass of 5.0 kg attached to a spring of stiffness 980 N/m. If the Co-efficient of friction is 0.025, calculate

- Frequency of free oscillations
- The no. of Cycles corresponding to 50% reduction in amplitude if the initial amplitude is 5.0 cm and
- the time taken to achieve 50% reduction.

Solution: For a horizontal system, the force is given as

$$F = \mu mg = 0.025 \times 5 \times 9.81 = 1.226 \text{ N}$$

(a) The natural frequency

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{980}{5}} = 14 \text{ rad/sec} = \underline{\underline{2.23 \text{ Hz}}}$$

(b) Amplitude after 50% reduction is half of the initial amplitude

$$\text{ie } \frac{5 \times 10^{-2}}{2} = 0.025 \text{ m}$$

Reduction in amplitude per cycle

$$= \frac{4F}{k} = \frac{4 \times 1.226}{980} = 5 \times 10^{-3} \text{ m}$$

Cycles Completed in 50% reduction.

$$\text{for } 1 \text{ Cycle} \rightarrow 5 \times 10^{-3} \text{ m}$$

$$\text{for } 'x' \text{ Cycles} \rightarrow 0.025$$

$$\therefore x = \frac{0.025}{0.005} = 5 \text{ cycles.}$$

(c) Time taken to achieve 50% reduction

$$= \underline{\underline{\text{no. of Cycles}}} \cdot \frac{2\pi}{\omega_n}$$

$$= 5 \times \frac{2\pi}{14} = \underline{\underline{2.22 \text{ sec}}}$$